

Test Cases:

Scenario 1: Start the process

```
PS D:\SEM-5\Distributed System\lab3> java Main.java 3
Process 0 with Unique ID 35 is initialized with local clock 0
Process 2 with Unique ID 37 is initialized with local clock 0
Process 1 with Unique ID 36 is initialized with local clock 0
```

Three processes are initialized, each with a logical clock value set to 0.

Scenario 2: Local events

```
local 0
-----
Process 0 local time 1
    Event type: LOCAL EVENT
    Event local time: 0
    Event content:
-----
local 1
-----
Process 1 local time 1
    Event type: LOCAL EVENT
    Event local time: 0
    Event content:
-----
```

Performing local events on Process 0 and Process 1 increments the logical clock of both processes by 1.

Scenario 3: Send and Receive events

<pre>send 0 2 ----- Process 0 local time 2 Event type: SEND EVENT Event local time: 0 Event content: ----- ----- Process 2 local time 3 Event type: RECEIVE EVENT Event local time: 2 Event content: -----</pre>	<pre>send 2 1 "Hi" ----- Process 2 local time 4 Event type: SEND EVENT Event local time: 0 Event content: ----- ----- Process 1 local time 5 Event type: RECEIVE EVENT Event local time: 4 Event content: "Hi" -----</pre>
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Process 0, with a local clock value of 1, sends a message to Process 2 with its logical clock value of 0. First, Process 0 increments its own clock by 1, updating it to 2. It then sends the message to Process 2 with the timestamp of 2. Upon receiving the message, Process 2 updates its clock to the maximum of its current value and the received timestamp, and then increments it by 1 which results in 3.

In second image, process can also send message alongside timestamp.

Scenario 4: Exit command

```
Local Time of process 0 is 2.  
Local Time of process 1 is 5.  
Local Time of process 2 is 4.
```

By using the `exit` command, the application returns a message indicating the final timestamp for each process.